

Software Requirements Specification (SRS) Template

Reference Template for Document Validation System

1. Introduction

1.1 Purpose

Describe the purpose of this Software Requirements Specification document. Mention the system name and explain what this document is intended to define.

1.2 Scope

Describe the scope of the system. Explain what the system will do and what the system will not cover.

1.3 Definitions, Acronyms, and Abbreviations

List important terms, acronyms, and abbreviations used in the document.

Term	Description
SRS	Software Requirements Specification
User	Person who interacts with the system
Admin	Person who manages the system

1.4 References

List any documents, websites, standards, or resources referred to when preparing this SRS.

1.5 Document Overview

Briefly explain how the rest of the document is organized.

2. Overall Description

2.1 Product Perspective

Explain how the system fits into the existing environment. Mention whether it is a new system, an improvement of an existing system, or part of a larger system.

2.2 Product Functions

List the main functions of the system.

- User registration and login
- Data submission
- Data validation
- Report generation
- Admin management

2.3 User Classes and Characteristics

Describe the different types of users who will use the system.

- Admin
- Registered User

- Guest User

2.4 Operating Environment

Describe the hardware, software, browser, operating system, or network environment required to run the system.

2.5 Design and Implementation Constraints

Mention any limitations or restrictions that affect the system design.

- The system must be web-based.
- The system must support PDF and DOCX uploads.
- The system must follow university/project guidelines.

2.6 Assumptions and Dependencies

List assumptions and dependencies related to the system.

- Users will have internet access.
- Uploaded documents will be in readable format.
- The system depends on a document parsing module.

3. System Features and Requirements

3.1 Functional Requirements

Describe the functions the system must perform.

FR-01: User Login

The system shall allow registered users to log in using valid credentials.

FR-02: Document Upload

The system shall allow users to upload documents in supported formats.

FR-03: Document Validation

The system shall validate uploaded documents against the selected document template.

FR-04: Validation Result Generation

The system shall generate a validation result showing missing sections, extra sections, structure score, and final status.

FR-05: Admin Template Management

The system shall allow admins to add, update, or remove validation templates.

4. External Interface Requirements

4.1 User Interface Requirements

Describe the user interface requirements of the system.

- The system shall provide a simple document upload interface.
- The system shall display validation results clearly.

- The system shall show errors if the uploaded file type is not supported.

4.2 Hardware Interface Requirements

Describe any hardware interfaces required by the system. If there are no special hardware requirements, mention that the system does not require special hardware.

4.3 Software Interface Requirements

Describe software components, APIs, databases, or external services used by the system.

- The system may use a database to store user records and validation results.
- The system may use a document extraction library to read PDF and DOCX files.

4.4 Communication Interface Requirements

Describe communication requirements such as internet connection, HTTP/HTTPS, email notifications, or API communication.

5. Non-Functional Requirements

5.1 Performance Requirements

Describe how fast or efficiently the system should work.

- The system shall validate an uploaded document within an acceptable response time.
- The system shall handle multiple document uploads without major performance issues.

5.2 Security Requirements

Describe security requirements of the system.

- The system shall allow only authorized users to access restricted features.
- Uploaded documents shall be stored securely.
- User passwords shall be encrypted or hashed.

5.3 Usability Requirements

Describe how easy the system should be to use.

- The system shall provide a user-friendly interface.
- The validation result shall be easy to understand.

5.4 Reliability Requirements

Describe how reliable the system should be.

- The system shall handle invalid files without crashing.
- The system shall provide meaningful error messages.

5.5 Availability Requirements

Describe the expected availability of the system.

- The system shall be available during working hours or according to organizational requirements.

5.6 Maintainability Requirements

Describe how easy it should be to maintain or update the system.

- The system shall allow new document templates to be added in the future.
- The system code shall be organized for easy maintenance.

6. Use Case Requirements

6.1 Use Case Diagram

Insert the use case diagram of the system.

6.2 Use Case Descriptions

Use Case ID: UC-01

Use Case Name: Upload Document

Actor: User

Description: The user uploads a document for validation.

Precondition: The user must be logged in.

Main Flow:

1. User selects the document upload option.
2. User chooses a document file.
3. System checks the file format.
4. System uploads the document.
5. System starts the validation process.

Postcondition: The document is uploaded successfully.

Use Case ID: UC-02

Use Case Name: Validate Document

Actor: System

Description: The system validates the uploaded document against the selected template.

Precondition: A document must be uploaded.

Main Flow:

6. System extracts text from the uploaded document.
7. System compares the document structure with the reference template.
8. System identifies missing and extra sections.
9. System calculates a structure score.
10. System generates the validation result.

Postcondition: The validation result is generated.

7. Data Requirements

7.1 Input Data

Describe the data entered or uploaded by users.

- User details
- Uploaded document
- Selected document type

7.2 Output Data

Describe the output generated by the system.

- Validation result
- Structure score
- Missing sections
- Extra sections
- Acceptance or rejection status

7.3 Database Requirements

Describe the data that needs to be stored in the database.

- User details
- Uploaded document details
- Validation results
- Template records

8. Validation Rules

8.1 Required Sections

The uploaded SRS document should include the following main sections:

- Introduction
- Overall Description
- System Features and Requirements
- External Interface Requirements
- Non-Functional Requirements
- Use Case Requirements
- Data Requirements
- Validation Rules
- Appendices

8.2 Validation Criteria

The document validation system should check:

- Whether required sections are available
- Whether section names are similar to the SRS template
- Whether sections follow the correct order

- Whether unrelated extra sections are included
- Whether the document belongs to the correct document type

8.3 Scoring Criteria

Score Range	Validation Result
75-100	Accept
50-74	Needs Improvement
0-49	Reject

9. Appendices

9.1 Appendix A: Sample Screens

Insert sample screen designs or wireframes if available.

9.2 Appendix B: Additional Information

Include any extra information related to the system.